

Department of Information Technology

University of Gujrat



Genetic-Insight

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IT- UOG - Project Management Office	Version: 1.0
Final Year Project Proposal Guide	Date: March 2, 2025

DECLARATION

I certify that project title **Genetic-Insight** is under my supervision with students **Ahtashanm-ul hassan, Syed Umer Zubair and Faria Mahboob** of Information Security, Faculty of Computing & Information Technology, University of Gujrat, Pakistan, worked under my supervision.

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1. Final Year Project Proposal

1.1 Introduction

This document outlines the preparation and submission of the final project proposal for “**Genetic-Insight**,” a SaaS platform designed to facilitate the classification of genes and proteins using machine learning models. **Genetic-Insight** will be a computationally intelligent system developed to support the research community working in the field of genomics and proteomics, enabling them to train several models to identify functions or behaviors from their compositional sequences.

The proposal will define the project's functional and non-functional requirements, scope, development timeline, processes, and justification for the chosen technologies. A professional and well-structured proposal will be presented under the following headings:

1.2 Project Title

Genetic-Insight - Machine Learning-Based platform served as a SAAS to support Classification of Genes and proteins

1.3 Project Overview statement:

Genetic-Insight is a cloud-based Software as a Service (SaaS) platform designed to streamline the classification of genes and proteins using advanced machine learning models. The platform empowers researchers and the scientific community to upload genetic or proteomic datasets and configure model settings through a user-friendly interface. Without requiring any programming skills, users can train, test, and evaluate machine learning models tailored to their specific data.

The goal of **Genetic-Insight** is to make complex machine learning tools accessible to the scientific community. By automating much of the process, the platform helps researchers save time, focus on their core research, and uncover important insights about gene functions and protein behavior, potentially contributing to advancements in areas like medicine and biotechnology.

Project Overview Statement Template

Project Title: Genetic-Insight																	
Project Manager: Mr. Junaid Muzaffar																	
Project Members:																	
Name	Registration #	Email Address	Signature														
Ahtasham-ul-hassan	21014156-001	21014156-001@uog.edu.pk															
Syed Umer Zubair	21014156-016	21014156-016@uog.edu.pk															
Faria Mahboob	21014156-024	21014156-024@uog.edu.pk															
Project Goal: The goal of Genetic-Insight is to develop a computationally intelligent system using machine learning models for the behavioral classification of protein and gene sequences. This platform aims to provide researchers with the ability to easily analyze genetic and proteomic data, identify patterns, and predict functions or behaviors of sequences. By offering a user-friendly, no-code solution, Genetic-Insight supports scientists in accelerating their research and making data-driven discoveries in fields such as genomics, proteomics, and biotechnology.																	
Objectives: <table border="1"> <tr> <th>Sr.#</th> <th>Objectives</th> </tr> <tr> <td>1</td> <td>Create a reliable and scalable system to process raw genetic sequences efficiently.</td> </tr> <tr> <td>2</td> <td>Implement various models for classifying protein and gene sequences.</td> </tr> <tr> <td>3</td> <td>Allow users to adjust model configurations for optimal performance.</td> </tr> <tr> <td>4</td> <td>Automated Process: Streamline training, testing, and evaluation of models for quick results.</td> </tr> <tr> <td>5</td> <td>Create an intuitive platform for training modules for users' own datasets and requirements.</td> </tr> <tr> <td>6</td> <td>Ensure the platform can handle large datasets and computational tasks.</td> </tr> </table>				Sr.#	Objectives	1	Create a reliable and scalable system to process raw genetic sequences efficiently.	2	Implement various models for classifying protein and gene sequences.	3	Allow users to adjust model configurations for optimal performance.	4	Automated Process: Streamline training, testing, and evaluation of models for quick results.	5	Create an intuitive platform for training modules for users' own datasets and requirements.	6	Ensure the platform can handle large datasets and computational tasks.
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Project Success criteria: Genetic-Insight will be measured by the following criteria: • High Classification Accuracy: Achieve a high level of accuracy in classifying protein and gene sequences. • Effective Feature Engineering: Implement robust feature engineering techniques that enhance model performance. • User-Friendliness: Ensure the platform is intuitive and easy to navigate for users of all skill levels. • Scalability and Reliability: The platform must effectively handle increasing data volumes and user loads without performance issues. • User Satisfaction: Attain a high user satisfaction rating through feedback and surveys. • Low Error Rate: Maintain a low error rate in model predictions to ensure reliability.																	

Assumptions, Risks and Obstacles:	
Assumptions	Risks
Users have basic knowledge of biological concepts.	Users may have difficulty understanding complex models.
Sufficient computational resources are available.	Resource limitations could hinder performance or scalability.
The platform will be compatible with various data formats.	Data incompatibility may lead to errors in processing.
Machine learning models used (e.g., Random Forest, SVM) are well-suited for the problem	Model selection may not yield optimal results, leading to subpar classification accuracy.
Users will provide high-quality, clean datasets.	Poor-quality data could affect model accuracy and reliability.
Machine learning models can effectively learn from the data provided.	Models may underperform if the data is insufficient or unrepresentative.
Obstacles: Development Time: The time required to develop and launch the platform may extend beyond initial estimates, affecting timelines and resources. Data Quality: Obtaining high-quality datasets is crucial; poor data can hinder model performance. User Adoption: Encouraging researchers to transition to a new platform may be challenging. Risk of model overfitting. Technical Complexity: Ensuring usability for non-technical users requires careful design and support. Handling class imbalance in datasets. Resource Limitations: Inadequate computational resources may restrict scalability and performance. Integration of ML models into a user-friendly platform. Integration Challenges: Technical difficulties in integrating with existing tools or databases. Regulatory Compliance: Navigating data privacy regulations can be complex and time-consuming.	
Organization Address: University Of Gujrat (Hafiz Hayat Campus)	
Type of project:	70 % Research 30% Development
Target End users: Researchers: Conducting studies in genomics and proteomics. Academic Institutions: Universities with biology and bioinformatics departments. Biotechnology Companies: Involved in drug discovery and genetic research. Healthcare Professionals: Seeking insights from genetic data for patient care. Bioinformaticians: Analyzing biological data using computational techniques. Pharmaceutical Companies: Identifying therapeutic targets in drug development. Government Bodies: Monitoring genetic data for public health compliance. Students: Learning and conducting projects in relevant fields.	
Platform:	☐ Web based
Approved By:	
Date:	

1.4 Project Goals & Objectives:

Project Goal:

Develop a computationally intelligent system using machine learning models for the behavioral classification of protein and gene sequences, enabling researchers to analyze genetic and proteomic data efficiently.

Objectives:

1. User-Friendly Interface:

Create an intuitive platform for training modules with users' own datasets and requirements.

2. Machine Learning Models:

Implement various models for classifying protein and gene sequences effectively.

3. Configurable Settings:

Allow users to adjust model configurations for optimal performance based on their needs.

4. Automated Process:

Streamline the training, testing, and evaluation of models to provide quick results.

5. Comprehensive Reports:

Deliver detailed analysis and visualizations of model predictions to facilitate user understanding.

6. Scalability:

Ensure the platform can handle large datasets and user loads efficiently.

7. Support Research:

Accelerate discoveries in genetics and proteomics by making advanced computational tools accessible to the scientific community.

1.5 High-level system components:

➤ **Input:**

Users can upload various types of datasets, including gene and protein sequences in formats like FASTA, CSV, or Other. The system will validate the data format and provide feedback on any issues.

➤ **Feature Engineering Module:**

Extracts relevant features from the input proteomics and genomics sequences, including: Nucleic Acid Composition (NAC), Di-Nucleotide Composition (DNC), Tri-Nucleotide Composition (TNC) etc.

➤ **Preprocessing and Feature Normalization:**

Techniques are applied to clean and standardize the data, ensuring consistency in the input for machine learning models. This may include scaling numerical features and encoding categorical variables.

➤ **Feature Selection and Optimization:**

This component identifies the most relevant features from the dataset that contribute to model performance. Techniques like Recursive Feature Elimination (RFE) or feature importance scores will be utilized.

➤ **Dataset Balancing Techniques:**

Implementing methods like oversampling or undersampling to address class imbalance in the dataset, ensuring that the models are trained effectively and do not favor one class over another.

➤ **Dataset Splitting:**

The system will split the dataset into training, validation, and test sets, typically following an 80-10-10 or 70-15-15 ratio, to evaluate model performance accurately.

➤ **Model Implementation:**

Various machine learning algorithms, such as Random Forest, Support Vector Machines (SVM), and Neural Networks, will be implemented for training and classification based on user-selected parameters.

➤ **Cross-Validation Techniques:**

Applying k-fold cross-validation to assess the robustness of the models and to minimize overfitting, ensuring that the models generalize well to unseen data.

➤ **Model's evaluation and Results Visualization:**

Providing users with visual representations of the model's performance metrics (e.g., confusion matrices, ROC curves) and predictions, along with comprehensive reports summarizing the analysis.

1.6 List of optional functional units:

➤ **Deep Learning:** Excludes complex deep learning models like CNNs or RNNs

➤ **Support for Image processing:**

Allow users to upload images and apply preprocessing techniques like resizing and normalization.

➤ **Image Upload:**

Allow users to upload images and apply preprocessing techniques like resizing and normalization.

➤ **Model training**

Use algorithms, such as Convolutional Neural Networks (CNNs), to automatically extract relevant features from images.

➤ **Image Classification Models:**

Offer pre-trained models (e.g., VGG16, ResNet) or options to train custom models for image classification.

➤ **Data Augmentation:**

Apply transformations (e.g., rotation, flipping) to enhance the training dataset.

➤ **Visualization of Results:**

Show visual predictions, such as bounding boxes or heatmaps, to illustrate model outputs.

➤ **Model Retraining:**

Enable users to retrain or fine-tune models with new images.

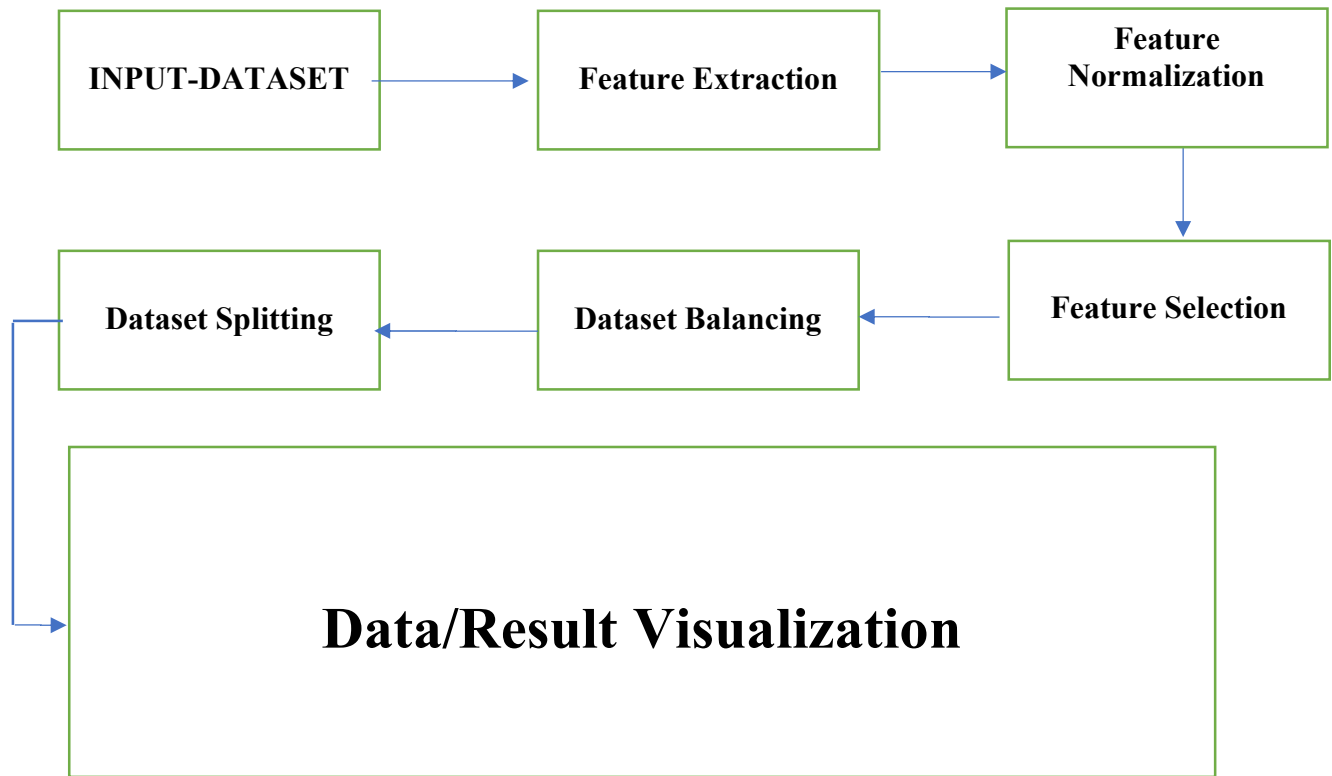
1.7 Exclusions:

- **Real-time Data Acquisition:** The platform will not include functionalities for real-time data collection from lab instruments or sensors.
- **Custom Algorithm Development:** The project will not support the creation of custom machine learning algorithms; it will utilize established models and methods.
- **Automated Data Collection:** No automated data scraping or real-time collection.
- **Distributed Systems:** No large-scale distributed system deployment.
- **Multi-language Support:** The platform will initially be developed in English, with no plans for multi-language support in the early phases.

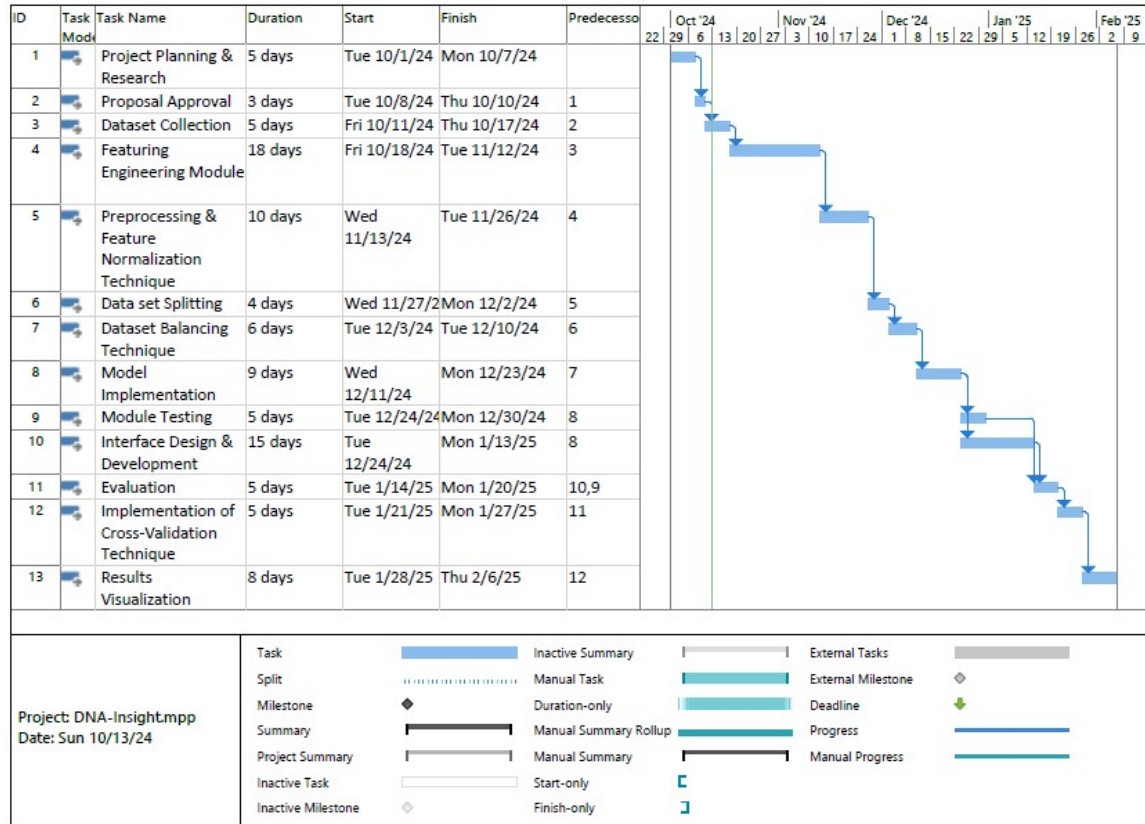
1.8 Application Architecture:

The architecture for **Genetic-Insight** will follow a modular approach, comprising the following components:

1. **Input Layer:** Accepts raw genetic sequences (DNA/RNA) from the user in various formats (e.g., FASTA, CSV).
2. **Machine Learning Classifier:** Utilizes various algorithms such as Support Vector Machines (SVM), Random Forest, and Neural Networks to classify the sequences based on extracted features.
3. **Output Layer:** Displays the classification results (e.g., epitope or non-epitope) and provides confidence scores for each prediction, enabling users to assess the reliability of the results.
4. **Validation Module:** Allows researchers to test and refine model accuracy using labeled datasets, ensuring that the models perform well with real-world data and can be adjusted based on feedback.



1.9 Gantt chart:



1.10 Hardware and Software Specification:

➤ Server Requirements:

1. Hardware:

- **Processor:** Multi-core CPU (Intel i7 or higher) or GPU (NVIDIA CUDA enabled for deep learning).
- **RAM:** Minimum 16 GB (recommended 32 GB for larger datasets).
- **Storage:** SSD with at least 500 GB (scalable based on data size).

2. Software:

- **Operating System:** Ubuntu 20.04 (or higher) or Windows Server.
- **Web Server:** Nginx or Apache (for serving the web application).
- **Programming Environment:** Python 3.8 or higher.
- **Machine Learning Libraries:** Scikit-learn, NumPy, Pandas.
- **Database:** MySQL or PostgreSQL.
- **Bioinformatics Tools:** BioPython.

➤ User Devices:

1. Hardware:

- **Processor:** Dual-core CPU (Intel i5 or equivalent).
- **RAM:** 4 GB or higher.
- **Storage:** 20 GB free space.
- **Display:** 1366 x 768 resolution or higher.

2. Software:

- **Operating System:** Windows 10, macOS, or Linux.
- **Browser:** Google Chrome, Firefox, or Safari (latest versions).
- **Additional Requirements:** Stable internet connection for web access.

1.11 Tools and technologies used with reasoning:

- **Jupyter Notebook:**

Reasoning: Jupyter Notebooks are excellent for rapid prototyping, testing, and visualizing results. It allows for interactive coding, which is useful for experimenting with various feature extraction techniques and algorithms.

- **NumPy and Pandas:**

Reasoning: These are essential for data manipulation, handling large genetic datasets, and performing operations like feature extraction (e.g., calculating nucleotide composition).

- **Matplotlib and Seaborn:**

Reasoning: Data visualization tools are crucial for plotting sequence properties, model performance metrics, and providing graphical insights into the classifier's results.

- **TensorFlow/PyTorch (Optional):**

Reasoning: These libraries will be used if the project decides to implement deep learning techniques (e.g., Convolutional Neural Networks) for sequence classification. They offer flexibility and performance optimizations, especially when working with GPUs.

- **Git/GitHub:**

Reasoning: Version control is necessary to manage code development, ensure collaboration, and track changes. GitHub also provides cloud storage for the project's codebase.

- **Docker (Optional):**

Reasoning: For containerization, ensuring that the project can be run in the same environment across different machines. This is particularly useful if there are multiple dependencies.